

Chp ~ 4

# d & f Block Elements

⇒ Position and General Introduction.

→ d block occupies the middle of the periodic table, Group 3-12 across Periods 4, 5, 6 and 7.

→ They are called Transition elements as they lie between p and s bridging transition between p and s.

⇒ Transition element is one whose atom in an oxidation state has an incompletely filled d-subshell (1-9 electrons in d-orbital)

→ In the d block elements Zn, Cd, Hg (Group 12) elements are not transition elements as they have filled d shell.

→ The requirement to become a transition element is the d orbital should be half filled (or not fully filled.).

Note: All Transition elements are d block elements, but not all d block elements are transition elements.

⇒ Three complete transition series.

| Series      | Elements                     | Period. |
|-------------|------------------------------|---------|
| 3d (First)  | Sc (21) → Zn (30)            | 4       |
| 4d (second) | Y (39) → Cd (48)             | 5       |
| 5d (Third)  | La (57) → Hg (80)<br>Hf (72) | 6       |

⇒ f block Elements.

It sits separately at bottom.

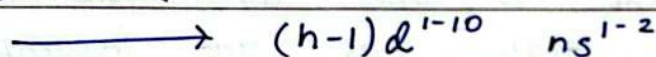
Group 3, periods 6, 7.

→ Lanthanoids : Ce (58) → Lu (71) following La (57)

→ Actinoids : Th (90) → Lr (103) following Ac (89)

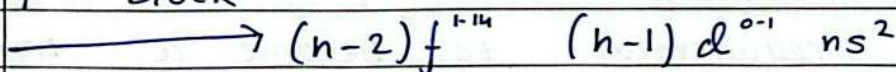
⇒ General Electronic Configuration.

# D-Block



| Element | Expected    | Actual         | Reason                            |
|---------|-------------|----------------|-----------------------------------|
| Cr (24) | $3d^4 4s^2$ | $3d^5 4s^1$    | Extra stability of half filled d  |
| Cu (29) | $3d^9 4s^2$ | $3d^{10} 4s^1$ | Extra stability of fully filled d |

# F-Block



- Lanthanoids : [Xe]  $4f^{1-14} 5d^{0-1} 6s^2$

- Actinoids : [Rn]  $5f^{1-14} 6d^{0-1} 7s^2$

## ⇒ Properties of Transition Elements.

i] Atomic Radii : Across a period

→ decrease initially and decrease becomes small after midway.

3d-4d → Normal increase

4d-5d → Barely changes due to Lanthanoid contraction

Down the group.

ii] Ionic Radii :

Since, transition metals exhibit various oxidation states. the radii of ions are different in different oxidation states.

iii] Metallic Character :

All are metals — hard, high tensile strength, good conductors, lustrous, malleable, ductile

Strong metallic bonding from both ns and (n-1)d electrons

iv] Ionisation enthalpy.

Ionisation enthalpies of d block are higher than that of s block and lower than p block.

It is due to increase in nuclear charge

The first ionisation enthalpies of 3d are greater than 1d and 2d.

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v] Oxidation States: Most of them have variable O.S. usually differing by 1.

Minimum O.S. is 0 + 2 and max is +8.

They appear highest with O and F.

→ The bonds formed by lesser O.S. are ionic and higher O.S. has covalent bond.

## vi] Formation of Coloured compounds. Ions

Most of the compounds of transition metals are coloured both in solid state as well as aqueous solution.

→ It is due to d-d transitions. Ligands split degenerate d-orbitals; an electron absorbs visible light to jump levels due to which complementary colour is observed.

→  $d^0$  ( $Sc^{3+}$ ,  $Ti^{4+}$ ) and  $d^{10}$  ( $Zn^{2+}$ ,  $Cu^+$ ) ions are colourless.

completely filled d-orbitals or completely empty d-orbitals are colourless.

## vii] Formation of complexes.

Transition metal ions form a large number of complex compounds.

They can form due to:

- i] Small Size
- ii] High Nuclear Charge
- iii] Available vacant d orbitals.

viii] Alloy Formation.

Transition metals form a number of alloys among themselves.

This tendency to form among themselves is due to almost similar atomic sizes. due to which metal can easily replace other metal from its lattice to form a solid solution called alloy.

⇒ Inner transition elements / f block elements.

→ Differentiating electrons enters an  $f$ -orbital of the ante penultimate  $(n-2)$  shell (inner transition).

→ placed separately at bottom, purely to keep the table compact.

i) Oxidation State : The most common Oxidation state of Lanthanoids is +3.

— A few show +2 or +4, only where it gives a specially stable  $f^0/f^7/f^n$  configuration.

ii) Formation of coloured compounds :  
Most of the actinoid ions are coloured. The ions having  $5f^0, 5f^1, 5f^7$  configuration are colourless. except these all are coloured.  $5f^3 \rightarrow$  red  $5f^2 \rightarrow$  green.

iii) Formation of complexes: As compared to d-block transitions, they only form few complexes. and that too less readily. due to

i) The size of lanthanoids is very large reducing force of attraction.

ii) In lanthanoids,  $f$ -orbitals are not available to form hybrid orbitals.

iv] Formation of alloys : Lanthanoids are highly dense. — they readily form alloys especially with iron. few important ones are  
 i] Misch metal  
 ii] Pyrophoric alloys.

v] Lanthanoid contraction: It has been observed that on moving from cerium to lutetium size of atoms and ions decrease.

→ It is the gradual decrease in atomic / ionic ( $M^{3+}$ ) radii from La (57) to Lu (71).

→ Cause:  
 each added electron enters the inner 4-f subshell which shields the rising nuclear charge very poorly (diffuse shape) effective nuclear charge on outer electron rises steadily → electron cloud pulled inward.

⇒ Consequences of Lanthanoid contraction.

i] 4d / 5d Similarity : contraction nearly cancels the expected size increase down a group.

ii] Difficulty separating lanthanoids from each other:  
 very gradual size change → very similar chemistry

iii] Basicity of oxides and hydroxides : Due to this the basic strength of oxides and hydroxides of lanthanoids decreases with increase in atomic number.

⇒ Chemical Reactivity with elements.

Lanthanoids are highly reactive metals, electropositive metals.

i) with oxygen: Lanthanoids tarnish readily on exposure to air resulting in the fading of their silver white lustre.

→ They burn in air to form their respective oxides.

ii) with hydrogen: They react to form non-stoichiometric hydrides. ( $\text{LnH}_2$ ) and ( $\text{LnH}_3$ )

iii) with halogens ( $\text{F}_2, \text{Cl}_2, \text{Br}_2, \text{I}_2$ ): they react to form trihalides, ( $\text{LnX}_3$ ) or ( $\text{LnX}_2$ ).

iv) with sulphur: they form sulphides ( $\text{Ln}_2\text{S}_3$ ).

v) with nitrogen: they react with nitrogen on warming to form nitrides.

vi) with carbon: they form ionic salt like carbides, and liberate hydrocarbons on hydrolysis.

vii) with water: they dissolve readily in warm water to liberate hydrogen.

✗ Actinoids.

Oxidation states:

+3 is again a characteristic state, but actinoids show a much greater range of oxidation states. till +7.

(U shows +3 to +6)

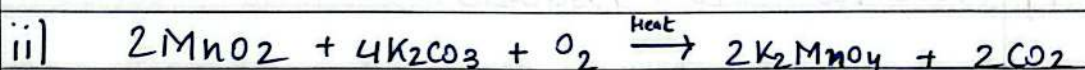
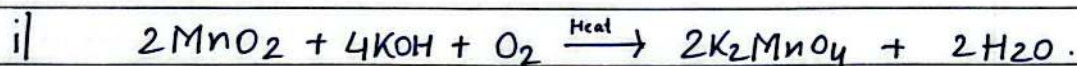
This is mainly due to the 5f, 6d and 7s orbitals are much closer in energy to one another than the 4f, 5d and 7s like lanthanoids.

## ⇒ Comparison between Lanthanoids and Actinoids.

| Feature                         | Lanthanoids                           | Actinoids.                                       |
|---------------------------------|---------------------------------------|--------------------------------------------------|
| Differentiating electron enters | 4f orbital                            | 5f orbital                                       |
| General config                  | $[\text{Xe}] 4f^{1-14} 5d^{0+1} 6s^2$ | $[\text{Rn}] 5f^{1-14} 6d^{0+1} 7s^2$            |
| Oxidation State                 | (+3) dominant                         | (+3) but 4, 5, 6 also common                     |
| Radioactivity                   | Non - Radioactive                     | All are radioactive                              |
| Availability                    | Naturally occurring in amounts        | Beyond (U <sub>92</sub> ) only in trace amounts. |
| Colour of ions                  | Generally pale / faint                | more intensely coloured                          |
| Complex forming                 | Weak tendency to form complex         | much greater tendency to form.                   |

## ~~XX~~ Potassium Permanganate ( $\text{KMnO}_4$ )

$\Rightarrow$  Preparation from pyrolusite Ore ( $\text{MnO}_2$ )



potassium  
permanganate

|                |
|----------------|
| Green<br>mass. |
|----------------|

$\Rightarrow$  Structure and Shape.

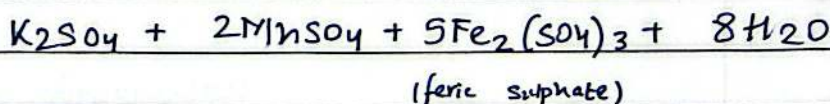
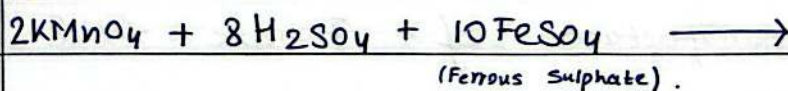
It is  $\text{sp}^3$  hybridized  
and has a tetrahedral structure.

## ~~XX~~ Oxidizing Nature.

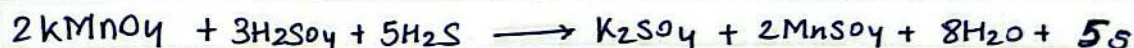
i) <sup>potassium</sup> (KI) Iodide to Iodine. (Basic)



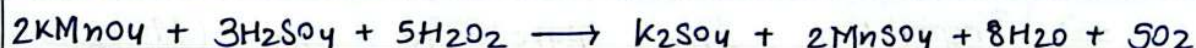
ii) ferrous salts to ferric salts. (Acidic)



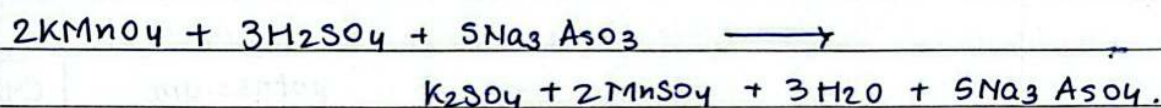
iii) Hydrogen sulphide to Sulphur (Neutral)



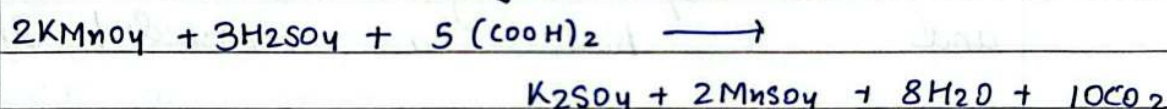
iv] Hydrogen peroxide ( $H_2O_2$ ) to water ( $H_2O$ ) [Acidic]



v] Arsenites to Arsenates.



vi] Oxalic Acid (oxalates) to Carbon dioxide.



⇒ Uses of  $KMnO_4$

- i] As an oxidising agent in volumetric (redox) titrations (estimation of  $Fe^{2+}$ , oxalate etc).
- ii] As a disinfectant / antiseptic in dilute solution.
- iii] For bleaching wool, cotton, silk and oils.
- iv] In manufacture of Benzoic Acid.

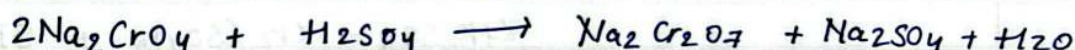
## ✗ Potassium Dichromate ( $K_2Cr_2O_7$ ).

⇒ Preparation from Chromite Ore.

i) Chromite ore to Sodium Chromate



\* ii) Chromate to Dichromate



iii) Sodium dichromate to Potassium dichromate

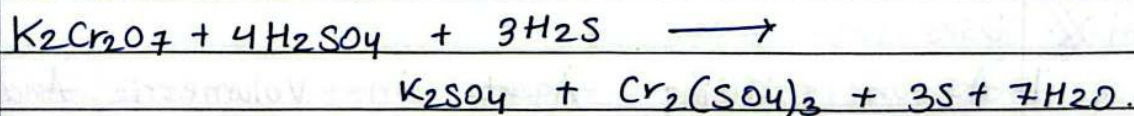
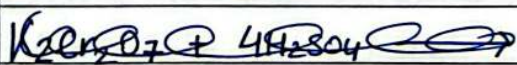


⇒ Structure and Shape.

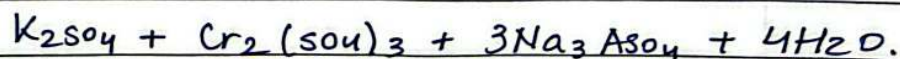
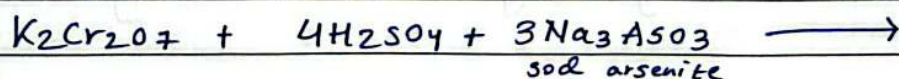
The dichromate  $Cr_2O_7^{2-}$  consists two tetrahedral  $CrO_4$  units sharing one oxygen corner and Cr is in +6 O.S.

⇒ Oxidising Nature.

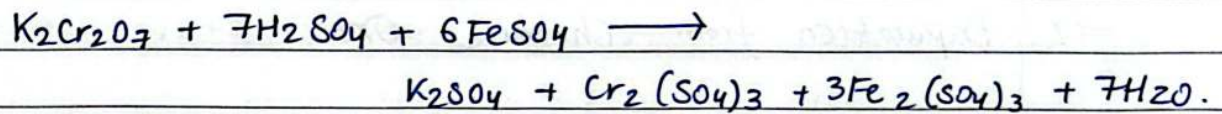
i) Hydrogen Sulphide to Sulphur.



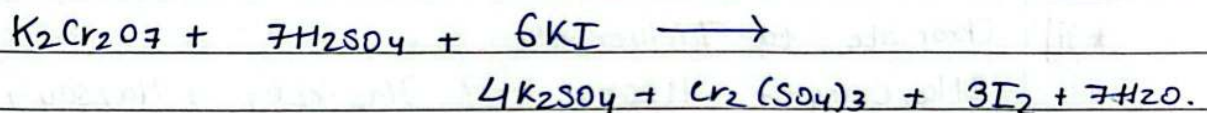
ii) Arsenites to Arsenates



iii] ferrous Salts to ferric Salts.



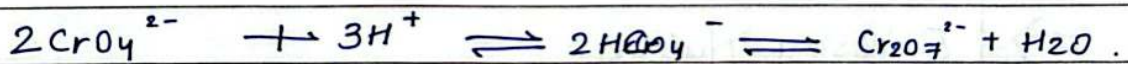
iv] Iodine from potassium Iodide



⇒ Uses of  $K_2Cr_2O_7$

i] Conversion of chromate ion to dichromate

\* the chromate and dichromate ion exist in equilibrium and thus are interconvertible by changing the pH.



Chromate  
ion - yellow

Hydrogen  
chromate

Dichromate  
ion.

✗ Uses

- i] As an oxidising Agent in Volumetric Analysis.
- ii] In Leather Industry for Chrome tanning
- iii] As an oxidiser in preparing many organic compounds.
- iv] In Laboratory glassware cleaning.

## ⇒ Give Reason Questions

Q.1] Enthalpies of Atomisation of transition metals is high.

Ans. Transition metals have strong metallic bonds between the atoms of these elements, thus high amount of heat is required to break metal lattice to give free atom.

Q.2] The transition metals generally form coloured compounds.

Ans. As they have incomplete d subshell or unpaired electrons in their d-subshell due to which d-d transition takes place and they become coloured.

Q.3] The transition metals and many of their compounds show paramagnetic behaviour.

Ans. As they have unpaired electrons in their  $(n-1)d$  shell.

Q.4] The lowest oxide of transition metal is basic the highest is acidic.

Ans. In the lower oxidation state of metals some of the valence electrons are not involved in bonding. Hence, it can donate electron and behave as a base but in higher oxidation states valence electrons are involved in bonding and are not available.  $\therefore$  It can accept electrons and hence, behave as an acid.

Q.5] Copper is considered as transition metal.

Ans. This is because (Cu) in oxidation states of (+2) has electronic configuration  $4s^0 3d^9$  that is incompletely filled d subshell.

Q.6] Chromium is a typical hard metal, but Mercury is a liquid.

Ans. Chromium has 5 unpaired 'd' electrons in the d subshell ( $4d^5 4s^1$ ). Hence, metallic bonds are very strong. In mercury all the d orbitals are fully filled  $3d^{10} 4s^0$ , hence metallic bond is very weak.

Q.7] Scandium is colourless and Titanium is coloured.

Ans. Scandium has no unpaired electrons but Titanium has 1 unpaired electron.

Q.8] Transition metals have high enthalpies of hydration.

Ans. Because they have small atomic size and high nuclear charge, both properties change as we move from left to right along the transition series.

Q.9] Transition elements show variable oxidation states

Ans. In these elements the energies of (n-1) d ~~electrons~~ orbitals and ns orbitals are very close, hence, electrons from both can take part in bonding.

Q.10] Transition elements have many irregularities in their electronic configuration.

Ans. Due to very small difference in the energy of  $(n-1)d$  subshell and  $ns$  subshell the incoming electron may enter into  $ns$  subshell or  $(n-1)d$  subshell.

Q.11] The paramagnetic character in  $3d$  series upto manganese and then decreases.

Ans. On moving from Scandium to manganese the no of unpaired electrons increases but after manganese the pairing of electrons start and no of unpaired electrons decrease.

Q.12] Zinc salts are white but  $Cu^{2+}$  salts are blue though both are  $d$  block.

Ans. Zinc has fully filled  $d$  subshell. while copper has half filled  $d$  subshell.